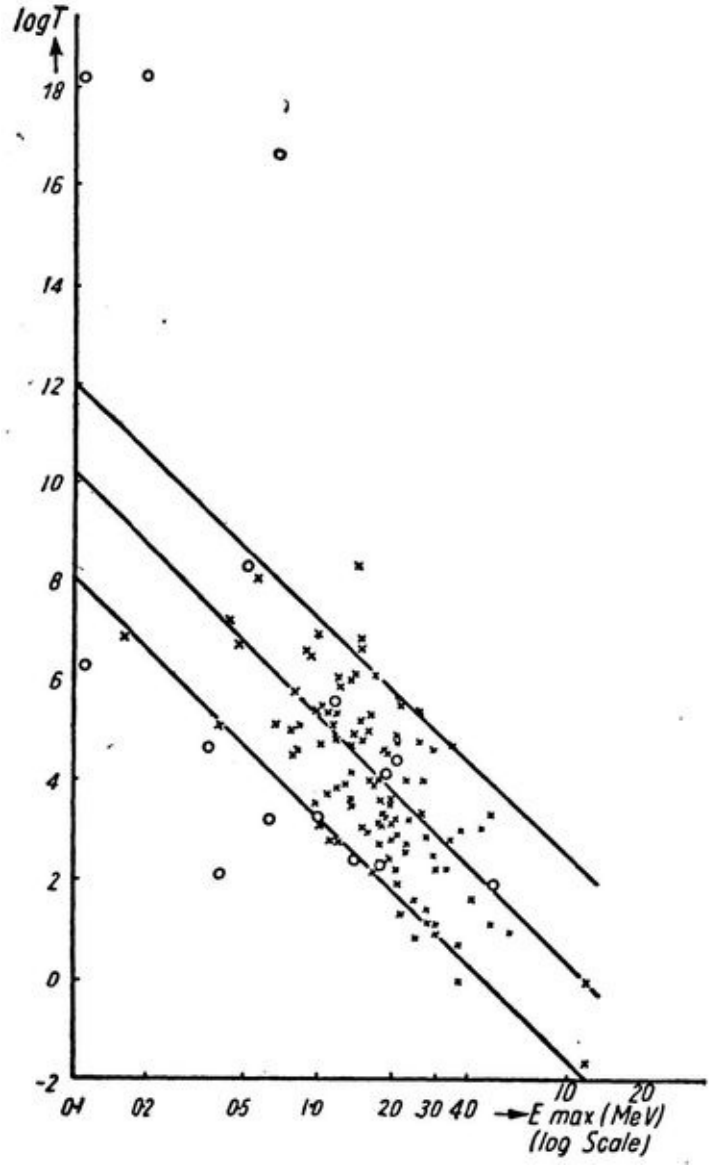


Figure 23.—Life and energy of natural and artificially produced beta radiators (*Sargent diagram*).

As always, the mass numbers and atomic numbers (superscripts and subscripts) must balance on both sides of the arrow.

The boron atom ${}_5B^{12}$, for instance, emits an electron and changes into the carbon atom ${}_6C^{12}$. Simultaneously with the emission of the electron, a neutrino is emitted, of which both the rest mass and the charge are 0; its symbol, therefore, is ${}_0\nu^0$. We write, consequently:

$$ {}_5B^{12} \rightarrow {}_6C^{12} + {}_{-1}e^0 + {}_0\nu^0 $$